

ALAMEIN INTERNATIONAL UNIVERSITY

Faculty of Computer Science & Engineering

Knowledge Based Systems | AIE212

Medical Diagnosis Expert System

Clinical Triage & Preliminary Diagnosis KBS

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PHASE 2: IDENTIFY SOLUTION

2.1 Proposed Knowledge-Based Approach

For this system, we decided to go with a Rule-Based Expert System (RBES) as the core approach, paired with a Certainty Factor (CF) mechanism. The CF idea comes from the MYCIN expert system, and it's specifically designed to deal with the kind of uncertainty that comes up all the time in medical diagnosis.

Primary Approach: Rule-Based Expert System (RBES)

The knowledge base stores medical expertise as IF–THEN rules. Each rule encodes a specific clinical reasoning step — basically the kind of logic a doctor goes through when they see a patient. Here's an example:

```
Rule R-08 (Red Flag – Suspected Acute MI) :  
IF   patient_age >= 40  
AND  symptom = chest_pain  
AND  pain_character = crushing_or_pressure  
AND  pain_radiation = left_arm OR jaw  
AND  symptom = diaphoresis (sweating)  
AND  spo2 < 95 OR heart_rate > 100  
THEN diagnosis = Acute_Myocardial_Infarction [CF = 0.95]  
AND urgency = IMMEDIATE  
AND  red_flag = TRUE  
AND  action = [call_emergency_services, aspirin_300mg]
```

Supplementary Approach: Certainty Factors (CF)

In practice, patients often don't present with textbook-perfect symptoms. Certainty Factors (between 0.0 and 1.0) are assigned to each rule to reflect how confident we are in a given conclusion. This lets the system:

- Consider multiple possible diagnoses at the same time and rank them by confidence.
- Let partial symptom matches still contribute to a conclusion, even if not every condition in a rule is met.
- Combine CF scores when several rules point to the same diagnosis, which gradually increases confidence the more evidence there is.
- Show the clinician how certain the system is, rather than giving a yes/no answer with no context.

2.2 Justification of the Rule-Based Approach

We chose RBES over other approaches like neural networks or Bayesian models for several reasons specific to the medical domain:

Justification Criterion	Why RBES Fits the Clinical Domain
Clinical Knowledge is Codifiable	Medical knowledge is already well-documented in guidelines from bodies like WHO, NICE, and ESC. It's the kind of structured, rule-like knowledge that translates directly into IF-THEN rules without much loss.
Transparency is Clinically Mandatory	Doctors and regulators need to understand why a system made a decision. RBES is naturally explainable — you can see exactly which rules fired. Machine learning models can't offer that level of transparency.
Established Medical Precedent	RBES has a long track record in medical AI. MYCIN in the 1970s and INTERNIST-1 are well-known examples that showed this approach actually works for clinical decision support.
No Large Dataset Required	Neural networks need thousands of labeled patient records to train on, which we don't have. With RBES, the knowledge comes directly from clinical guidelines and expert input — much more realistic for a university project.
Uncertainty is Manageable	Medical cases are often ambiguous, but not infinitely so. CFs handle that uncertainty well enough without going into full Bayesian probability networks, which would be much more complex to implement.
Modifiability & Maintenance	If medical guidelines are updated, you just add or change the relevant rules. The rules are readable enough that a domain expert could do it without needing to understand the underlying code.

2.3 Type of Reasoning

The system uses a combination of two reasoning strategies — forward chaining and backward chaining — and switches between them depending on the situation:

Forward Chaining (Data-Driven) — Primary Mode

This is the main mode. It mirrors how a normal clinical intake works — starting from what the patient reports and working toward a conclusion. The inference engine takes the inputs, matches them against rules, fires the matching ones, and keeps going until it reaches a diagnosis and urgency level.

- **When used:** Used in every standard assessment, where the person just reports what they're experiencing and we work forward from there.
- **Process:** All inputs go into Working Memory as facts. The engine matches those against rule conditions, fires the rules that match, adds the conclusions back into Working Memory as new facts, and repeats until nothing new can be derived.
- **Clinical example:** Patient reports chest pain + sweating + left arm pain → Rule R-08 fires → AMI hypothesis added to WM with CF=0.95 → Red flag rule fires → Urgency set to IMMEDIATE → Action rules fire → Output generated.

Backward Chaining (Goal-Driven) — Verification Mode

Backward chaining is used when a clinician already has a specific diagnosis in mind and wants to confirm or rule it out. It's also used internally when the system wants to proactively check for dangerous conditions like stroke.

- **When used:** Used when a clinician wants to verify a specific diagnosis, or when the system automatically checks for high-priority conditions after detecting certain symptoms (e.g., triggering a stroke check if neurological symptoms are reported).
- **Process:** The system starts from the target diagnosis, works out which rules are needed to confirm it, and asks for any missing data — so it only requests what's actually needed rather than going through the whole intake again.
- **Clinical example:** System suspects stroke based on slurred speech → Backward chaining checks: Is there facial drooping? Arm weakness? Sudden onset? → Prompts clinician for any missing observations → Confirms or rules out stroke hypothesis.

Justification for Hybrid Reasoning

- Forward chaining covers the typical case where no diagnosis is assumed yet — which is what happens in most real assessments.
- Backward chaining is there for when a clinician suspects something specific, or when the system needs to actively verify a dangerous condition before moving on.
- Together, the two modes reflect how doctors actually think: they observe and collect data (forward), then mentally check off serious possibilities before settling on a conclusion (backward).

2.4 Conceptual Decision Flow: From Input to Recommendation

From the moment a patient's data is entered to the final output, the system goes through nine steps:

Step	Description
1 — Patient Data Collection	The patient or clinician fills in a structured form covering symptoms, vitals, medical background, and history. Inputs are checked for completeness and valid ranges (e.g., SpO2 has to be between 50–100%).
2 — Fact Assertion	Every validated input becomes a fact in the Working Memory (e.g., FACT: symptom=chest_pain, FACT: spo2=93, FACT: age=58, FACT: known_condition=hypertension). This is what the inference engine works from.
3 — Red Flag Pre-Check	Before any standard diagnosis begins, the system does a quick scan for immediately dangerous patterns — things like a suspected heart attack, stroke, anaphylaxis, or respiratory failure. If any of those are detected, it jumps straight to IMMEDIATE urgency and the correct emergency action, skipping the normal diagnostic flow.

4 — Rule Matching (Agenda)	The pattern matcher checks all rules against the current facts in Working Memory. Any rule whose conditions are satisfied gets added to the Agenda — a prioritized queue of rules ready to fire.
5 — Conflict Resolution	When multiple rules match at the same time, the system resolves conflicts using three criteria: (a) more specific rules win over general ones, (b) rules with higher CF scores are preferred, and (c) rules pointing to higher urgency are promoted. Patient safety always takes priority.
6 — Rule Firing & Derivation	The top-priority rule fires and its conclusion is added to Working Memory as a new fact. This can trigger more rules, and the cycle keeps going until no new facts can be derived.
7 — Differential Consolidation	All diagnostic hypotheses that were derived are ranked by their total CF score. The highest-scoring one becomes the primary diagnosis; up to three others are listed as alternatives with their confidence levels.
8 — Action & Urgency Assignment	Recommendation rules fire based on the confirmed diagnosis and urgency level, producing a specific action plan that takes into account the care setting (GP clinic, telemedicine, or emergency department).
9 — Explanation & Output	The explanation module traces which rules fired and writes a plain-language summary of the reasoning. The full output — diagnosis, differentials, urgency level, recommended actions, red flag alerts, and reasoning — is then shown to the clinician.